# Autonomous Physics-Based Modeling of Retention Loss in Cryogenic 3-D NAND Flash

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***Abstract*— A novel model is established with physical derivation in retention operation of 3-D NAND flash memory. To efficiently incorporate the retention loss properties of the immediate and continuous occurance, an autonomous model has been suggested. By utilizing the model, retention loss for both short-term and long-term due to detrapping effect of trapped electron can be simulated in very short timeframe. Moreover, autonomous model makes it possible for model to predict the time transient of $\Delta V_{th}$ by only using an initial condition.**

***Index Terms*— Autonomous modeling, detrapping, 3-D NAND flash memory, retention loss modeling**

## I. Introduction

THERE rapid increase in data storage demand, driven by emerging computing technologies such as artificial intelligence (AI) and high-performance computing (HPC), has raised significant concerns about the reliability of 3-D NAND flash memories, particularly regarding retention loss [1].

As storage capacity per cell increases, ensuring long-term stability of each bit's threshold voltage distribution over time scales from days to years becomes increasingly critical. Moreover, prior works have shown that charge detrapping can still occur at cryogenic temperatures, supporting earlier models that treat detrapping as largely temperature independent [2].

In this study, an autonomous model for retention loss based on tunneling emission mechanisms in 3-D NAND flash memory is developed. The proposed simulation framework requires only the initial condition as input and enables highly efficient computation with significantly reduced simulation time.

## II. Modeling

This section presents the modeling framework for autonomous simulation of retention loss. First, a cylindrical Poisson solution is derived for the oxide–nitride–oxide (O/N/O) gate stack, incorporating quantized multiple layers within the charge trap nitride (CTN) layer. After obtaining the detailed Poisson solution for these discretized layers, the threshold voltage shift ($\Delta V_{\mathrm{th}}$) is calculated using an optimized numerical scheme. Finally, based on previous studies of detrapping-induced emission rates [3], an autonomous equation governing retention loss behavior is proposed.

### A. Detailed Poisson Solution

An electrical potential distribution obtained from the cylindrical Poisson equation for discretized layers in the ONO structure can be expressed as follows [3], [4].

$$\begin{cases} V_{\mathrm{Si}}(r_{\mathrm{Si}}) = V_{\mathrm{ch}} \\ V_{\mathrm{TOX}}(r) = V_{\mathrm{ch}} + \frac{K_{\mathrm{TOX}}}{\epsilon_{\mathrm{TOX}}} \ln \frac{r}{r_{\mathrm{Si}}} \\ V_{\mathrm{CTN},i}(r) = V_{\mathrm{CTN},i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{\mathrm{CTN}}} \ln \frac{r}{r_{i-1}} \\ V_{\mathrm{BOX}}(r) = V_{\mathrm{CTN},N}(r_N) + \frac{K_{\mathrm{BOX}}}{\epsilon_{\mathrm{BOX}}} \ln \frac{r}{r_N} \\ V_{\mathrm{BOX}}(r_{\mathrm{BOX}}) = V_{\mathrm{G}} \end{cases} \tag{1}$$

where $i = 1$–$N$ denotes the index of each discretized slice of the CTN layer in the radial direction, and $N$ is the total number of radial partitions.

Solving $\mathbf{x}$ vector can calculate the total $N+2$ potential parameters.

$$\mathbf{x} = \begin{bmatrix} K_{TOX} & K_1 & \cdots & K_N & K_{BOX} \end{bmatrix}^\top \tag{2}$$

$$A\mathbf{x} = \mathbf{b} \tag{3}$$

Matrix A and vector $\mathbf{b}$ are constructed by enforcing a total of $N+1$ transverse electric field continuity conditions ($\epsilon_1 \mathcal{E}_1 = \epsilon_2 \mathcal{E}_2$) together with an additional boundary condition defined by reference potential.

$$A = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{\mathrm{TOX}}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\mathrm{BOX}}} \end{bmatrix} \tag{4}$$

$$\mathbf{b} = \begin{bmatrix} \frac{\Delta Q_0}{2\pi} & \frac{\Delta Q_1}{2\pi} & \cdots & \frac{\Delta Q_N}{2\pi} & \Delta V \end{bmatrix}^\top \tag{5}$$

$$\Delta Q_i = \pi r_i^2 \left( n_{t,i} - n_{t,i-1} \right) \tag{6}$$

$$\Delta V = (V_{\mathrm{G}} - V_{\mathrm{ch}}) - \sum_{i=1}^{N} \frac{q n_{t,i}}{4 \epsilon_{\mathrm{CTN}}} \left( r_i^2 - r_{i-1}^2 \right) \tag{7}$$

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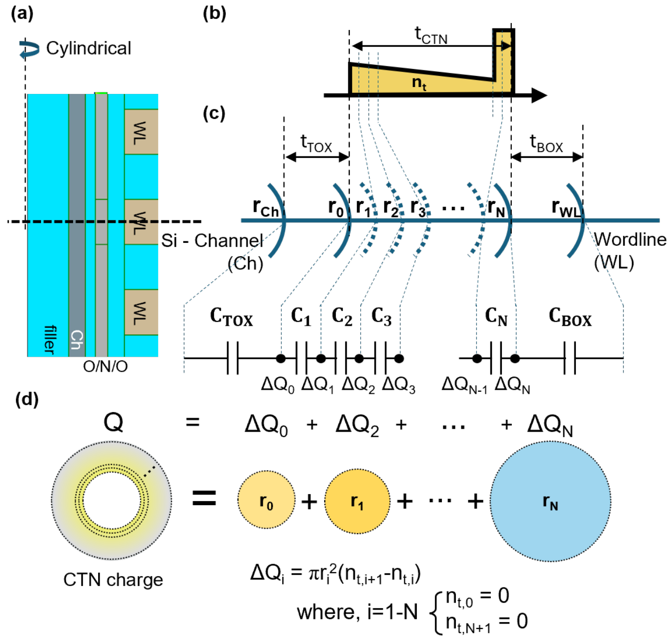



Fig. 1. (a) shows a radial scale and capacitance visulization, (b) shows the representation of $\Delta Q_i$ within the boundary condition solution, (c) shows the programming sequence of trapped electrons and holes, and (d) shows the autonomous simulation schematics.

$C_{\mathrm{TOX}}$, $C_i$, and $C_{\mathrm{BOX}}$ in (4) denote the capacitances of each cylindrical layer, as illustrated in Fig. 1(c). $\Delta Q_i$ in (5) represents the charge component associated with the cylindrical shell at radius $r_i$, which can be superpositioned as $Q = \sum \Delta Q_i$ as demonstrated on Fig. 1(d). Furthermore, (5) indicates that $\mathbf{b}(\Delta Q_i, V_G - V_{\mathrm{ch}})$ varies with the trapped charge distribution and the applied voltage conditions.

## B. Q - $\Delta V_{th}$ Relation

For the $\Delta V_{\mathrm{th}}$ calculation, the gate voltage difference required to maintain equal channel field conditions must be determined [3], [4].

$$F_{\mathrm{TOX}}(r_{\mathrm{Si}}) = \frac{K_{\mathrm{TOX}}}{\epsilon_{\mathrm{TOX}}} \frac{1}{r_{\mathrm{Si}}} \tag{8}$$

To achieve equal channel field conditions, the same $K_{\mathrm{TOX}}$ value under the given conditions must be established. By Cramer's rule, the $K_{\mathrm{TOX}}$ parameter can be solved through determinant calculation.

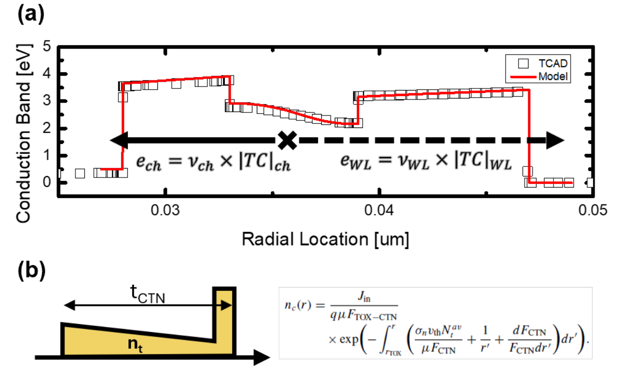
$$K_{\mathrm{TOX}} = \frac{\det\left[A_{\mathbf{b}(\Delta Q_i, \Delta V_{\mathrm{th}})}\right]}{\det A} \tag{9}$$

where notation $A_{\mathbf{b}(\Delta Q_i, \Delta V_{\mathrm{th}})}$ stands for $1^{\mathrm{st}}$ column of matrix $A$ replaced with vector $\mathbf{b}(\Delta Q_i, \Delta V_{\mathrm{th}})$.

Therefore, $\Delta V_{\mathrm{th}}$ can be solved easily by using the property of multilinearity.

$$\det\left[A_{\mathbf{b}(0, \Delta V_{\mathrm{th}})}\right] = -\Delta V_{\mathrm{th}} = \det\left[A_{\mathbf{b}(\Delta Q_i, 0)}\right] \tag{10}$$

$$\therefore \Delta V_{\mathrm{th}} = -\det\left[A_{\mathbf{b}(\Delta Q_i, 0)}\right] \tag{11}$$




(c) Other Parameters

| μ [eV] | σ [eV] | $N_t$ [$cm^{-3}$] |
|---|---|---|
| 0.84 | 0.2 | $1.15 \times 10^{19}$ |
| 1.8 | 0.05 | $3.5 \times 10^{19}$ |

Fig. 2. (a) The TCAD simulation calibration with the measurement from previous study [2]. (b) A detail detrapping curve variation over time with $t_{\mathrm{TOX}}$ from 3 nm to 5 nm.

The threshold voltage difference can be calculated with single determinant computation with initially programmed trapped charge distribution is given as Fig. 1(b).

## C. Autonomous Detrapping Modeling

With the given initial conditions, an autonomous equation is established for each cell, divided by radial position and trap depth as shown in Fig. 2(b).

$$\frac{d}{dt} n_t(r, E_t, t) = -(e_{\mathrm{ch}} + e_{\mathrm{WL}}) n_t(r, E_t, t) \tag{12}$$

Then, detrapping relation can be express as following general ODE solution.

$$n_t(r, E_t, t) = \sum_r \sum_{E_t} n_{t0}(r, E_t) \exp(-(e_{\mathrm{ch}} + e_{\mathrm{WL}})t) \tag{13}$$

The emission rate for trapped electron and hole can be computed as following equations based on a previous study [3].

$$e = \nu \times |TC| \tag{14}$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\mathrm{trap}} \cdot \frac{m_0 \sqrt{2m^*}}{\pi^2 \hbar^3} \sqrt{E - E_{\mathrm{trap}}} \tag{15}$$

$$|TC| = \prod \exp\left(-\frac{4\sqrt{2m^*}}{3\hbar q} \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F}\right) \tag{16}$$

for both channel and wordline directions as demonstrated in Fig. 2(b).

This autonomous detrapping modeling methodology enables a simulation to directly comput the result at a specific time without performing any transient time-interval calculations. Only the initial condition and the target time are required, eliminating the need for time-stepped simulations.

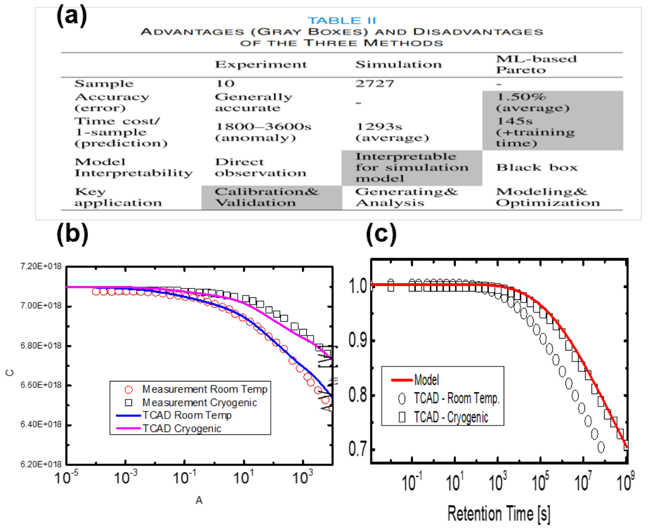

(a)

TABLE II
ADVANTAGES (GRAY BOXES) AND DISADVANTAGES OF THE THREE METHODS

| | Experiment | Simulation | ML-based Pareto |
|---|---|---|---|
| Sample | 10 | 2727 | - |
| Accuracy (error) | Generally accurate | - | 1.50% (average) |
| Time cost/ 1-sample (prediction) | 1800–3600s (anomaly) | 1293s (average) | 145s (+training time) |
| Model Interpretability | Direct observation | Interpretable for simulation model | Black box |
| Key application | Calibration& Validation | Generating& Analysis | Modeling& Optimization |



Fig. 3. (a) The TCAD simulation calibration with the measurement from previous study [2]. (b) A detail detrapping curve variation over time with $t_{\mathrm{TOX}}$ from 3 nm to 5 nm.

## III. SIMULATION AND RESULT

According to Refaldi et.al [2], cryogenic conditions enable separation of the detrapping mechanism. To emulate cryogenic conditions during retention operation within the TCAD framework, the capture cross-section was adjusted to an extremely low value ($\sigma_{\mathrm{n}} \rightarrow 0$) only during retention. This manipulation suppresses thermal electron transitions between trapped and conduction states within the CTN layer, as illustrated in Fig. 2(a), while maintaining physical consistency in other operations.

The model validation is performed using Synopsys TCAD simulations calibrated against previous retention modeling studies [3], [5]. To properly focus on the detrapping effect from the CTN layer, bandgap-engineered processes were neglected and program operation was perfomed with fresh cell. The simulation was perfomed with virgin cell with program voltage of 1 $\mu s$ applied to the target wordline. To ignore lateral migration retention loss, neighboring data pattern was selected [9].

The variation of the $t_{ox}$ showed the most dominancy in detrapping time transient curve. In Fig. 2 (b), $t_{ox}$ from 3 nm to 5 nm for both TCAD simulation and the model were perfomed, showing good match with each other.

## IV. CONCLUSION

This study presents an autonomous model for retention loss due to detrapping of trapped electrons. Compared to TCAD simulations calibrated with physical parameters from various prior studies, the proposed model demonstrates good match with TCAD simulation. Additionally, the novelty of autonomous modeling is demonstrated through significant computational optimization, achieving results within an extremely short timeframe.

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